# Source coding

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## Codes

**Definition 0.1** (Binary code)**.** Let $X$ be a discrete random variable. A **binary code** for $X$ assigns each value $x$ a *codeword*, a finite string of binary digits, of length $\ell(x)$.

**Definition 0.2** (Uniquely decodable code)**.** A binary code is **uniquely decodable** if every finite sequence of codewords, written one after another, can be split back into codewords in only one way.

**Definition 0.3** (Expected length)**.** The **expected length** of a binary code for a discrete random variable $X$ is the expectation of its codeword length:

$$L \stackrel{\text{def}}{=} \mathbb{E}[\ell(X)].$$

**Theorem 0.1** (Expected length as a sum over values)**.** *Let $X$ be a discrete random variable with probability mass function $p(x) \stackrel{def}{=} \mathbb{P}(X = x)$. The expected length of a binary code for $X$ is*

$$L = \sum_x p(x)\,\ell(x).$$

Proof

*Proof.* The codeword length $\ell(X)$ is a random variable whose possible values are the codeword lengths $\ell(x)$. By the definition of expectation,

$$L = \sum_k k\,\mathbb{P}(\ell(X) = k).$$

The event $\ell(X) = k$ is the union of the disjoint events $X = x$ over the values $x$ with $\ell(x) = k$. So $\mathbb{P}(\ell(X) = k) = \sum_{x:\ell(x)=k} p(x)$, and

$$L = \sum_{k} \sum_{x:\ell(x)=k} k\, p(x).$$

On the inner sum $k = \ell(x)$. Every value $x$ lies in exactly one inner sum, so the double sum runs once over each value:

$$L = \sum_{x} \ell(x)\, p(x).$$

□

**Example 0.1** (Expected length of a code for a three-valued variable)**.**
Let $X$ have probability mass function $p = (1/2, 1/4, 1/4)$, and let a binary code for $X$ have codeword lengths $\ell = (1, 2, 2)$. The code below computes the expected length from its definition, as the expectation of $\ell(X)$, and as the sum in Theorem 0.1.

```
p <- c(1 / 2, 1 / 4, 1 / 4)
len <- c(1, 2, 2)
lengths_seen <- sort(unique(len))
prob_of_length <- vapply(
  lengths_seen,
  function(k) sum(p[len == k]),
  numeric(1)
)
from_definition <- sum(lengths_seen * prob_of_length)
from_sum <- sum(p * len)
c(from_definition = from_definition, from_sum = from_sum)
#> from_definition        from_sum
#>             1.5             1.5
```

Both give $L = 1.5$ bits, as Theorem 0.1 states.

*Remark* 0.1 (Short codewords for common values). A code with a small expected length gives frequent values short codewords and rare values long ones, since each value's length counts in $L$ in proportion to its probability.

## The source coding theorem

**Theorem 0.2** (Source coding theorem)**.** *Let $X$ be a discrete random variable that takes finitely many values, at least two of them with positive probability, and let $H_2(X)$ be its entropy with base-2 logarithms.*

1. *Every uniquely decodable binary code for $X$ has expected length $L \geq H_2(X)$.*
2. *Some uniquely decodable binary code for $X$ has expected length $L < H_2(X) + 1$.*
3. *Coding blocks of $n$ independent copies of $X$ at a time, some uniquely decodable binary code spends, on average, fewer than $H_2(X) + 1/n$ binary digits per copy.*

*Remark* 0.2 (Where the theorem comes from). Shannon (1948) proved that entropy is the limit on how far the output of a source can be compressed. Cover and Thomas (2006, sec. 5.3 to 5.5) prove each part of Theorem 0.2. For part 2, they give each value $x$ with $p(x) > 0$ the length $\ell(x) = \lceil -\log_2 p(x) \rceil$, its surprisal in bits rounded up; the Kraft inequality (Cover and Thomas 2006, sec. 5.2) guarantees that a code with these lengths exists.

## Coding with the wrong distribution

**Theorem 0.3** (The cost of coding with the wrong distribution)**.** *Let $X$ take finitely many values, with probability mass function $p$, and let $q$ be a probability mass function, possibly different from $p$, with $q(x) > 0$ wherever $p(x) > 0$. Suppose a code gives each value $x$ with $p(x) > 0$ the length $\ell(x) = \lceil -\log_2 q(x) \rceil$. Then its expected length $L$ under $p$ satisfies*

$$H_2(p, q) \leq L < H_2(p, q) + 1,$$

*where $H_2(p, q)$ is the cross-entropy of $q$ relative to $p$, with base-2 logarithms.*

Proof

*Proof.* Every sum here runs over the values $x$ with $p(x) > 0$. Rounding up adds less than one binary digit, so for each such $x$

$$-\log_2 q(x) \leq \ell(x) < -\log_2 q(x) + 1.$$

Multiply each part by $p(x) > 0$ and sum. Since $\sum_x p(x) = 1$,

$$-\sum_x p(x)\log_2 q(x) \le \sum_x p(x)\,\ell(x) < -\sum_x p(x)\log_2 q(x) + 1.$$

The middle sum is $L$ by Theorem 0.1, and the outer sums are $H_2(p,q)$. □

**Theorem 0.4** (Cross-entropy is entropy plus divergence)**.** *Let $p$ and $q$ be probability mass functions on the values $x_1, \dots, x_n$, with $q(x) > 0$ wherever $p(x) > 0$. Write $H_2(p)$ for the entropy of $p$, $H_2(p,q)$ for the cross-entropy of $q$ relative to $p$, and $D_2(p \,\|\, q)$ for the Kullback–Leibler divergence of $q$ from $p$, all with base-2 logarithms. Then*

$$H_2(p,q) = H_2(p) + D_2(p \,\|\, q).$$

Proof

*Proof.* Every sum here runs over the values $x$ with $p(x) > 0$. For every such $x$,

$$-\log_2 q(x) = -\log_2 p(x) + \log_2 \frac{p(x)}{q(x)}.$$

Multiply by $p(x)$ and add over $x$:

$$-\sum_x p(x)\log_2 q(x) = -\sum_x p(x)\log_2 p(x) + \sum_x p(x)\log_2 \frac{p(x)}{q(x)}.$$

Reading left to right, these are $H_2(p,q)$, $H_2(p)$ and $D_2(p \,\|\, q)$. □

**Example 0.2** (Cross-entropy, entropy and divergence for a uniform code)**.**
Let $p = (1/2, 1/4, 1/4)$ be the probability mass function of a three-valued random variable, and let $q = (1/3, 1/3, 1/3)$ assign $1/3$ to each of the three values. The code below computes the cross-entropy, the entropy and the divergence in bits.

```r
p <- c(1 / 2, 1 / 4, 1 / 4)
q <- rep(1 / 3, 3)
cross_entropy <- -sum(p * log2(q))
shannon_entropy <- -sum(p * log2(p))
divergence <- sum(p * log2(p / q))
c(
  cross_entropy = cross_entropy,
  shannon_entropy = shannon_entropy,
  divergence = divergence
)
#>   cross_entropy shannon_entropy      divergence
#>       1.5849625       1.5000000       0.0849625
```

The cross-entropy is $H_2(p, q) = 1.585$ bits. The entropy is $H_2(p) = 1.5$ bits, and the divergence is $D_2(p \parallel q) = 0.085$ bits. They satisfy Theorem 0.4: $1.5 + 0.085 = 1.585$.

*Remark* 0.3 (Cross-entropy is the cost of the wrong code). Theorem 0.4 splits the cross-entropy into the entropy and the divergence. By Theorem 0.2, no uniquely decodable code averages fewer than $H_2(p)$ bits. By Theorem 0.3, a code built for $q$ costs about $D_2(p \parallel q)$ bits per value more than that. By the divergence theorem, the excess vanishes only when the code was built for $p$ (Cover and Thomas 2006, sec. 2.6; Yeung 2008, sec. 2.5). Cover and Thomas (2006, sec. 5.4) state Theorem 0.3 in this form, with $H_2(p) + D_2(p \parallel q)$ in place of $H_2(p, q)$.

*Remark* 0.4 (Why models are fit by minimizing cross-entropy). Fitting a classifier by minimizing cross-entropy chooses the model's distribution $q$ so that a code built for $q$ would spend as few bits as possible on the observed labels. By Remark 0.3, that pushes $q$ toward the distribution the labels came from.
For a single labeled example, let $e$ be the distribution that puts all its mass on the observed label. Then $H_2(e) = 0$ (as for a coin that always lands heads), so the example's cross-entropy loss $H_2(e, q)$ equals $D_2(e \parallel q)$: every bit the loss reports is a wasted one.

## References


Cover, Thomas M., and Joy A. Thomas. 2006. *Elements of Information Theory*. 2nd ed. Wiley-Interscience. https://doi.org/10.1002/047174882X.

Shannon, C. E. 1948. "A Mathematical Theory of Communication." *Bell System Technical Journal* 27 (3): 379–423. https://doi.org/10.1002/j.1538-7305.1948.tb01338.x.

Yeung, Raymond W. 2008. *Information Theory and Network Coding*. Springer. https://doi.org/10.1007/978-0-387-79234-7.